# Janadri Yalla Yashwanth


**Software Engineer | B.Tech CSE (AI & ML) 2025 | Python · Java · SQL · Full-Stack Development**

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## SUMMARY


B.Tech in Computer Science (AI & ML) 2025 graduate, CGPA 8.6, with hands-on software development experience across full-stack web applications, Python/FastAPI backends, and AI/ML — currently building and supporting production systems at Marut Drones. Strong CS fundamentals (Data Structures & Algorithms, DBMS, OS) and end-to-end experience across development, testing, and deployment. Immediate joiner seeking a Systems Engineer role to build, customize, and implement enterprise products.


## TECHNICAL SKILLS

**Languages:** Python, Java, C, SQL

**CS Fundamentals:** Data Structures & Algorithms, DBMS, Operating Systems, OOP, REST API design

**Web / Frameworks:** FastAPI, React, HTML, CSS, Uvicorn (ASGI), async/await

**Databases:** SQL, PostgreSQL, Redis

**Tools / DevOps:** Git, Docker, Docker Compose, GitHub Actions CI/CD, Linux

**AI / ML (bonus):** PyTorch, OpenCV, TensorFlow, scikit-learn

**Tools:** Git & GitHub, Claude Code, VS Code, Docker, GitHub Actions, Postman, Linux

## EXPERIENCE

**Software Engineer Trainee — Marut Drones** *Feb 2026 – Present*

- Software engineer on the drone AI/automation team — shipped production computer-vision detection services, geospatial backend features, and autonomous-flight tooling; selected projects below.

**Machine Learning Intern — Rooman Technologies** *Jan 2025 – Mar 2025*

- Built ML solutions for healthcare; developed and evaluated a sleep-disorder classification model on physiological and behavioral data, achieving 98% accuracy.

## KEY PROJECTS — MARUT DRONES

**Flight Log Analyzer — Full-Stack Drone Diagnostics Platform**

*React, TypeScript, FastAPI, Python, Docker*

- Built a full-stack web app that ingests raw drone flight logs and auto-generates professional PDF health/crash reports, supporting both ArduPilot (.bin) and JIYI K++ formats with automatic binary-vs-text detection via magic-byte sniffing.
- Engineered a from-scratch binary parser for the ArduPilot/DataFlash self-describing log format using Python struct, decoding 50+ message types and reconstructing real-world IST timestamps from GPS-week data.
- Implemented a phased diagnostic engine running 50+ automated checks (power, IMU/vibration, GPS, EKF, compass, RC, failsafes), classifying findings as PASS/WARN/FAIL and inferring probable crash cause.
- Developed automated PDF reporting with ReportLab + Matplotlib (flight-path maps, sensor charts, color-coded tables); built the React 19 + Vite + Tailwind front end with file upload and download handling.
- Containerized the full stack with Docker / Docker Compose; integrated a .NET converter via Mono + Xvfb inside Linux (~7,200 LOC, solo-built).

**Marut Survey Platform — Backend & AI Detection Services**

*FastAPI, Celery, Redis, PostgreSQL, PyTorch, SAM-2, S3/GDAL, Docker*

- Built three production detection pipelines (stockpile, tree-crown, building) integrating three model families — SAM-2, DeepForest, and GroundingDINO/LangSAM — behind a FastAPI backend with async Celery/Redis job APIs (live progress, cancellation, owner-scoped RBAC).
- Derived per-object stockpile volumes (m3) and material classification from nDSM height integration, backed by a 7-factor confidence rubric and Union-Find cross-tile NMS dedup at 0.3 IoU.
- Streamed cloud-optimized GeoTIFFs block-by-block from S3 via GDAL /vsis3/ + rasterio (no full-raster load) and tuned S3 multipart transfers — 64 MB chunks, 25 concurrent threads, pooled connections — for multi-gigabyte orthomosaics.
- Packaged the PyTorch / SAM-2 / DeepForest / GroundingDINO stack into a Docker image shipped through GitHub Actions to systemd-managed GPU and CPU workers; Pydantic v2 schemas and Alembic migrations.

**Aerial Image Object Detection — Multi-Model Benchmark**

*Python, PyTorch, Hugging Face Transformers, SegFormer, OpenCV*

- Developed a computer-vision pipeline detecting buildings, vehicles, and trees from high-resolution drone imagery (stitched panoramas up to 14,400 x 7,200 px) using SegFormer-B5 (ADE20K, 150 classes) with PyTorch and Hugging Face Transformers.
- Benchmarked 4+ segmentation models (SegFormer-B5/B2, Aerial-Drone SegFormer, SAM + CLIP) head-to-head and selected ADE20K for best precision, cutting false positives ~3x versus over-segmenting alternatives.
- Engineered the full inference pipeline in OpenCV/NumPy: sliding-window tiling, HSV sky-skipping, semantic-mask-to-instance contour extraction, geometric shape filtering, and per-class Non-Max Suppression.
- Delivered annotated outputs (per-class count overlays + JSON), batch-folder processing, and drone-video support (frame-skip to annotated MP4), running on GPU/CUDA at ~7–11 s per image.

## EDUCATION

**Sri Venkateswara College of Engineering, Tirupati** *Dec 2021 – May 2025*
B.Tech in Computer Science (Artificial Intelligence & Machine Learning) — CGPA: 8.6

**Narayana Junior College, Hyderabad** *Jun 2019 – Mar 2021*
Intermediate (MPC) — 96%

## CERTIFICATIONS

- CS50's Introduction to Programming with Python — Harvard University Online
- Databases: Topics in SQL — Stanford Online